SPM-MIM: A super-pixel based masked image modeling method for multitemporal remote sensing land cover classification

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***Abstract*—In this paper, we propose a super-pixel based multitemporal masked image modeling (SPM-MIM) method for multitemporal remote sensing land cover classification. Compared with the previous self-supervised masked image modeling methods, SPM-MIM masks and reconstructs images based on super-pixels, effectively avoiding the problem that tiny remote sensing scenes are completely covered, while better retaining the structural information of remote sensing scenes, so that the model can be trained with stronger feature extraction capabilities. In order to better combine multi-temporal information to achieve land cover classification, we propose a cross-temporal information complementary module (CTICM). CTICM interacts with shallow multi-temporal features in spatial dimension to promote cross-phase information complementarity in local scenes, and interacts with deep multi-temporal features in channel dimension to promote cross-phase information complementarity in global scenes. Experimental results on remote sensing images of Gaofen-1 Harbin area and Gaofen-2 Dalian Lushun area show that our proposed SPM-MIM method has excellent land cover classification capability.**

Keywords-self-supervised learning; super-pixel masking; multitemporal remote sensing; remote sensing classification

I. INTRODUCTION

Land cover classification plays a pivotal role in the exploration and understanding of our planet. Remote sensing, which periodically acquires extensive ground object information, is widely used for large-scale classification [1][2]. Currently, most deep learning-based land cover classification models are designed for single-temporal remote sensing images (STRSI). When using such models, multi-temporal remote sensing images (MTRSI) must be classified separately, resulting in the loss of correlation information between different temporal phases. Therefore, there is a need for deep learning methods that can integrate multitemporal features for effective classification.

Noureddine [3] attempted to combine multitemporal information using a combination of SVM and genetic algorithms. However, this approach is less effective than deep learning-based single-phase methods because it is not data-driven. Another approach [4] is to concatenate STRSI in the channel dimension, but this requires substantial memory and introduces excessive redundant information. Additionally, high-quality labeled remote sensing images are scarce compared to the remote sensing image data. To mitigate the reliance on labeled images, self-supervised methods such as contrastive learning (CL)[5][6] and masked image modeling (MIM)[7][8] have been employed in STRSI processing to enhance the model's ability to extract global and general features. Unlike CL, MIM, a generative self-supervision method, excels at training models to perceive local remote sensing scenes. However, the fixed-size mask patches used in MIM can cover entire targets or scenes in remote sensing images, leading to significant structural information loss due to the prevalence of tiny targets or scenes.

To effectively combine multitemporal information for improved land cover classification, we draw inspiration from Changer[9], a remote sensing image change detection model. Changer's feature interaction module not only extracts image features but also enhances the extraction of bitemporal correlation information. However, this module primarily focuses on extracting change information. For land cover classification, it is crucial to utilize information from unchanged ground objects over multiple temporal periods to aid classification. Therefore, we build upon Changer by designing a cross-phase information complementarity module(CTICM). CTICM complements extracted shallow remote sensing scene features in the spatial dimension and deep features in the channel dimension, enabling the effective integration of multi-temporal information. To address the issue of small remote sensing scenes being obscured and losing overall structural information when using fixed patch blocks, we propose a masking strategy based on super-pixels. A super-pixel is a small region composed of adjacent pixels with similar features such as color, brightness, and texture. Typically, an object is divided into several super-pixels. Masking super-pixels avoids covering entire remote sensing scenes while preserving richer structural information. Building on this strategy, we propose a super-pixel based multitemporal masked image modeling (SPM-MIM) method for multitemporal remote sensing land cover classification.

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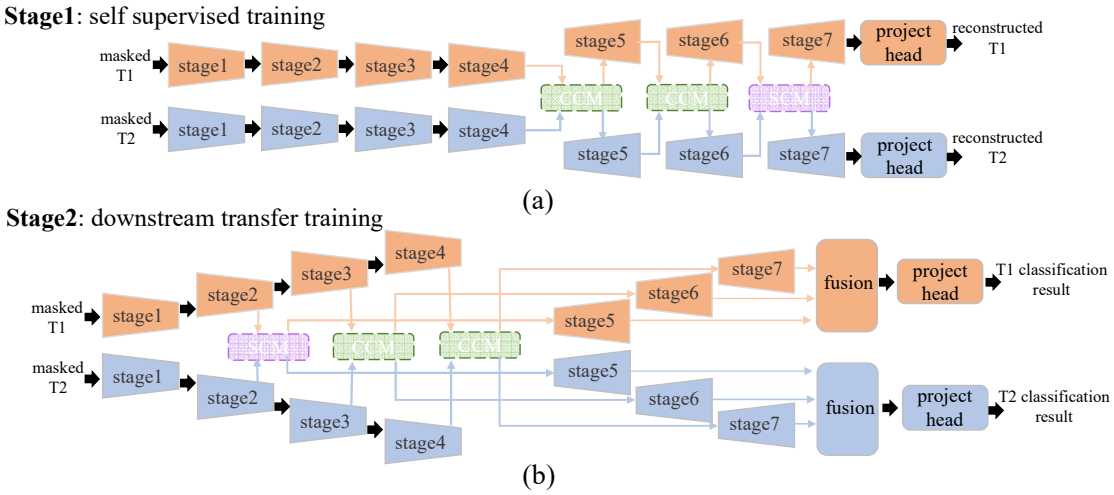


Figure 1. Overview of SPM-MIM method. (a) Model of self-supervised training stage. (b) Model of the downstream transfer training stage.

The contributions of this paper can be summarized as follows:

1) We propose a self-supervised land cover classification method called SPM-MIM. Unlike traditional masking strategies with fixed patch sizes, the super-pixel-based masking in SPM-MIM avoids covering entire tiny remote sensing scenes, thereby retaining more abundant structural information.

2) We introduce the CTICM to achieve land cover classification by integrating multi-temporal remote sensing information. CTICM enhances shallow multi-temporal features in the spatial dimension and deep features in the channel dimension through an attention mechanism, enabling models to more effectively utilize multi-temporal information.

II. METHODOLOGY

The overview of the SPM-MIM is shown Fig. 1. Similar to most MIM methods, the proposed SPM-MIM follows an encoder-decoder structure. However, since the input is bitemporal images, it needs to be adjusted to twin network architecture. This architecture comprises two sets of encoder and decoder with identical structure and parameters. Moreover, the CTICM is incorporated between the twin networks to realize information complementation across time features in space and channel dimensions.

The training of the SPM-MIM method is structured into two stages: self-training and transfer training. Since super-pixel segmentation does not participate in model training, it is performed separately on remote sensing images before model training. In the self-training stage, the model first applies a masking operation to the remote sensing image based on the super-pixel map, and then inputs the masked image into the encoder for feature extraction. Subsequently, these features are passed to the decoder for feature decoding. The CTICM module facilitates information complementarity in both spatial and channel dimensions, ultimately reconstructing the original image. This mechanism of using partial pixel information to reconstruct the original image allows the model to develop robust image feature extraction capabilities without the need for labeled samples.

Upon completing the self-training stage, the model's encoder attains strong remote sensing image feature extraction capabilities. However, this alone is insufficient for land cover classification. Therefore, it is necessary to perform transfer training on a set of labeled samples to tailor the model for this specific task. Given the differences between the downstream classification task and the self-supervised reconstruction task, we made targeted modifications to the decoder while retaining the weights of the encoder and CTICM. During the transfer training stage, remote sensing images are directly input into the encode without the masking operation for feature extraction. The extracted features are then passed to the decoder for feature decoding. Concurrently, the CTICM module facilitates information complementarity in both spatial and channel dimensions. This process enables effective land cover classification over multiple temporal phases. Upon completing the transfer training stage, the model demonstrates excellent land cover classification capabilities.

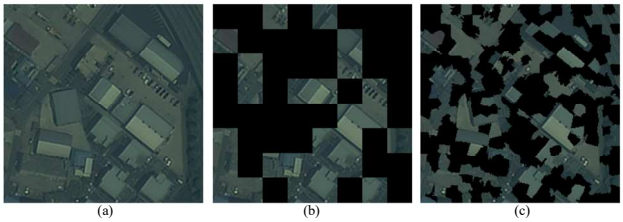


Figure 2. Comparison of mask between fixed patch and super-pixel. (a) The original image. (b) Fixed patch masked image. (c) Super-pixel masked image.

A. *Super pixel mask strategy*

In MIM, images are masked according to a specified patch size, and the model is trained to reconstruct these masked images, thereby enhancing its ability to extract image features. Currently, MIM is predominantly trained on image datasets in the computer vision (CV) field, where individual images typically contain few objects, and these objects occupy a significant portion of the image. Consequently, it is uncommon for mask blocks to cover entire objects or cause substantial loss of structural information. In contrast, remote sensing images exhibit

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significant variability in object sizes, including many small objects. Using a fixed-size mask block can result in the loss of critical scene information in remote sensing images, ultimately degrading the model's training effectiveness. To address this issue, we propose a masking strategy based on super-pixels.

First, the entire remote sensing image is segmented into a series of small regions composed of super-pixels with similar features such as color, brightness, and texture using a super-pixel segmentation algorithm. Specifically, we employ the SLIC [10] algorithm, known for its high efficiency and remarkable partitioning effect. Subsequently, super-pixels are randomly selected, masked, and their pixel values set to zero. As illustrated in Figure 2, the super-pixel based masking operation prevents the complete coverage of tiny remote sensing scenes and better preserves the structural information of the remote sensing scene.

B. *Cross-temporal information complementary module*

The CTICM module is inspired by Sheng's Changer, which demonstrates that feature interaction enhances a model's ability to perceive changing information, promotes the similarity of feature distributions between branches, and facilitates automatic domain adaptation across dual time domains. Since multitemporal image classification requires combining correlation information from different phases, we modified the original interaction operation to an information complementary operation. This involves complementing deep features in the channel dimension and shallow features in the spatial dimension. To enable the self-supervised training model to adapt to other tasks, we transferred the CTICM from the encoder to the decoder. In Changer, 50% of spatial dimension elements or channels are randomly selected for exchange, which can disrupt feature continuity and result in the exchange of unrelated features, thereby hindering model training. To address this issue, we introduced an attention mechanism.

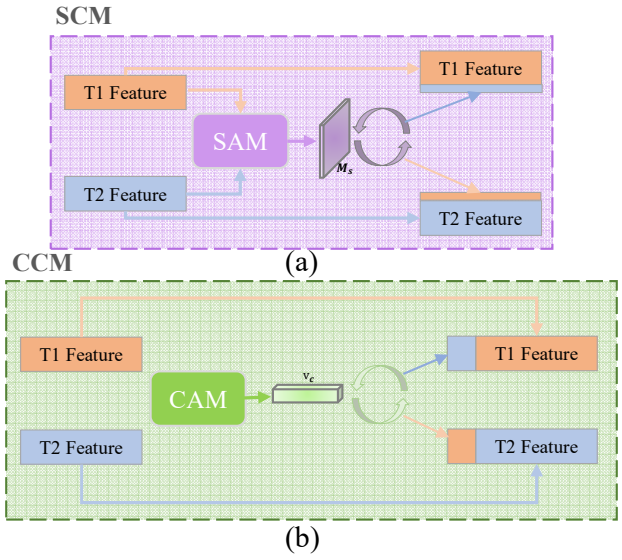


Figure 3. Overview of CTICM submodules. (a) Spatial complementary module. (b) Channel complementary module

As shown in Figure 3, the CTICM module comprises two sub-modules: the spatial complementary module (SCM) and the channel complementary module (CCM). SCM utilizes features extracted from bitemporal images to calculate a spatial attention map $\boldsymbol{M}_s \in \mathbb{R}^{1\times H\times W}$, which is then used to extract complementary features from the bitemporal image features. These complementary features are spliced into the original features. Similarly, CCM calculates a channel attention vector $\boldsymbol{v}_c \in \mathbb{R}^{C\times 1\times 1}$ based on the features of the bitemporal images, extracts complementary features according to $\boldsymbol{v}_c$, and splices them into the original features.

C. *Model training*

1) *Self training stage*

The self-training stage of SPM-MIM is illustrated at the top of Figure 1. The twin network structure shares weights. The encoder used here is the Transformer encoder from Segformer [11], which can also be replaced with encoders from ViT, Swin, and other models. The decoder integrates the CTICM, designed in this paper, into the MAE [12] model decoder.

$$\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1} = decoder(CTICM(encoder(\boldsymbol{F}_1^{l}, \boldsymbol{F}_2^{l}))) \quad (1)$$

where F_1^l, F_2^l represents phase i image features obtained at stage l. $i \in (1,2), l \in (1,2,3,4), \boldsymbol{F}_i^l \in \mathbb{R}^{C\times H\times W}$. Encoder consists of $LN(\cdot), TransformerBlock(\cdot)$ and $PatchEmbed(\cdot)$, and decoder consists of $LN(g), TransformerBlock(g)$ and $Upsample(\cdot)$.

In the self training stage, MSE loss was used to predict the loss of masked patches between the image and the original image.

$$\boldsymbol{L}_{sr} = \frac{\sum_{i,j}[(\boldsymbol{X}_i^{sr} - \boldsymbol{X}_i)\cdot \boldsymbol{T}_{mask}])}{\sum_{i,j}\boldsymbol{T}_{mask}} \quad (2)$$

2) *transfer training stage*

The transfer training stage of SPM-MIM is illustrated at the bottom of Figure 1. Self-supervised training primarily focuses on pretext tasks, such as reconstructing original images, rather than on classification tasks typical of downstream applications. Therefore, it is necessary to modify the decoder and incorporate a projection head specifically designed for classification tasks. The weights obtained from the previous training for the encoder and CTICM are retained.

$$\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1} = decoder(concat(CTICM(encoder(\boldsymbol{F}_1^{l}, \boldsymbol{F}_2^{l})))) \quad (3)$$

where $concat(\cdot)$ represents concatenate operation.

For the land cover classification downstream task, the conventional cross entropy is used here to calculate the loss:

$$L_{ce}(\boldsymbol{Y}, \boldsymbol{P}) = -\frac{1}{N}\sum_{i=1}^{N}[\boldsymbol{Y}_i log_2 \boldsymbol{P}_i + (1-\boldsymbol{Y}_i) log_2 (1-\boldsymbol{P}_i)] \quad (4)$$

III. EXPERIENCE

A. *Data description*

The data sets used in this experiment are the remote sensing images of Gaofen-1 Harbin area and Gaofen-2 Dalian Lushun area. The remote sensing image of the Harbin area of Gaofen-1 consists of four bands of R/G/B/NIR, including images on September 26, 2013 and September 16, 2014, with a resolution of 8m/pixel and a

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size of 450×2934, including vegetation, water, roads, crops, buildings and bare soil. The remote sensing image of the Dalian Lushun area of Gaofen-2 is composed of four bands of R/G/B/NIR, including images on September 6, 2015, June 28, 2016 and August 21, 2016, with a resolution of 4m/pixel and a size of 1536×1536. Includes vegetation, water, roads, crops, residential areas, bare soil, plant 1, plant 2, plant 3 these categories.

B. Experience setting

The proposed method was implemented using the PyTorch framework, and the model's training and testing were conducted on an NVIDIA RTX 4090 GPU. The super-pixel size for the masking operation was set to 8-16, with a masking rate of 35% during the self-supervised training stage. Throughout all stages of model training, the AdamW optimizer was employed, featuring a momentum of 0.9, a weight decay of 5e-4, and a learning rate of 5e-4. The self-supervised training stage was trained for 500 epochs, followed by 500 epochs for the downstream classification transfer training stage. To quantitatively evaluate the methods, this paper employs three evaluation metrics: overall accuracy (OA), individual class accuracy(CA) and kappa coefficient(Kappa).

In the self-training stage, two phases of GF1 and three phases of GF2 images are used as training data. In the transfer training stage, labeled data from 10% of the GF1 region is used for training, while the remaining 90% is used for validation. To verify the effectiveness of the proposed method, models without the super-pixel mask strategy and without the CTICM module are used as baselines. These baseline models are then compared to the proposed models with the respective designs incorporated.

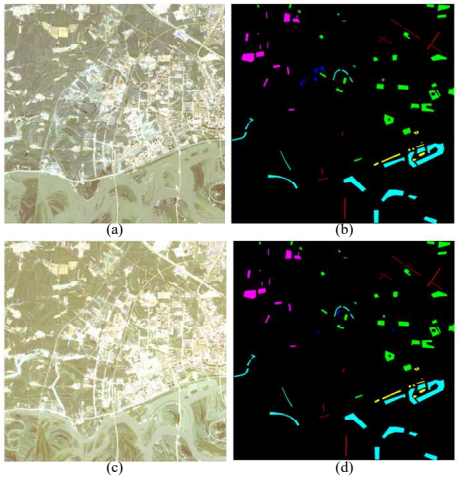


Figure 4. Gaofen-1 remote sensing images. (a) Gaofen-1 image dated 26 September 2013. (b) Gaofen-1 label dated 26 September 2013. (c) Gaofen-1 image dated 16 September 2014. (d) Gaofen-1 label dated 16 September 2014.

TABLE I. EXPERIENCE RESULT ON GAOFEN-1 DATA(%)

Class	Baseline	Baseline + super-pixel	Baseline + super-pixel + CTICM
Road	40.62	61.76	74.89
Building	75.66	86.34	92.17
Bare soil	40.04	66.40	75.62
Vegetation	69.23	77.07	82.48
Water	77.09	87.26	93.65
Crops	74.24	83.55	87.82
OA	94.52	98.78	99.03
Kappa	70.62	85.22	88.49

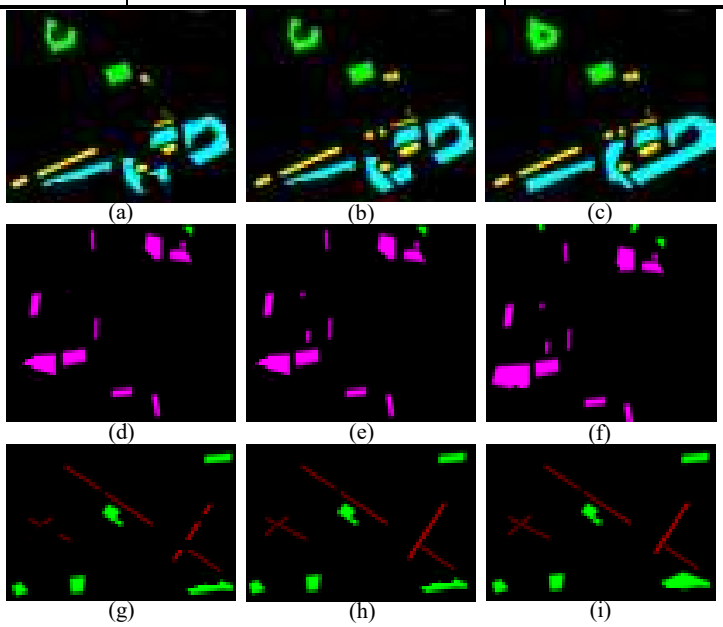


Figure 5. Local display of classification results. (a)(d)(g) is the baseline result. (b)(e)(h) is the result of adding the super pixel mask policy. (c)(f)(i) is the result of the addition of CTICM

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C. Experience result

As shown in Table 1, the baseline model demonstrates suboptimal classification performance, particularly for categories with abundant data such as waters and roads. This suggests that the model's ability to perceive remote sensing scenes is limited. With the introduction of the super-pixel mask strategy, there is a slight overall improvement in accuracy, and a significant enhancement in classifying micro-remote sensing scene categories such as bare soil and roads. This indicates that the super-pixel mask strategy effectively enhances the model's ability to perceive micro-scenes. Furthermore, the addition of the CTICM module results in a substantial improvement in overall accuracy, demonstrating that complementary information from multiple temporal phases can significantly enhance the model's classification performance.

IV. Conclusion

In this paper, we propose the SPM-MIM method for multitemporal remote sensing image land cover classification. Compared to the traditional MIM method with fixed patch blocks, SPM-MIM utilizes a super-pixel mask strategy to prevent complete coverage of tiny remote sensing scenes, thereby preserving the structural information more effectively. Additionally, the CTICM module we designed enables simultaneous feature complementarity in both spatial and channel dimensions, resulting in improved classification performance. Following self-supervised pre-training and supervised transfer training, the model demonstrates strong performance on GF1 data, achieving excellent classification results.

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